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Roll No

EX-801 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Computer Aided Electrical Machine Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. With the help of flow chart, explain Hooke and Jeeves method to solve unconstrained optimization problem.
2. a) Elaborate the method of choosing average gap density and choice of ampere conductor per metre from output equation of a synchronous machine.
b) How the variables can be selected for optimal design of an Alternator?
3. Discuss the various optimization problems and their mathematical formulations.
4. Discuss in detail various factors in the design of transformers for
 - i) Minimum cost
 - ii) Maximum efficiency
5. a) What are the factors which decide the selection of variables for optimal design of 3phase alternator? Explain.
b) Draw a flow chart for optimal design of armature winding of 3 ϕ alternative.
6. a) Explain the design procedure for design of armature winding of d.c. machine.
b) Calculate the length and diameter of the armature of a 7.5 kW, 220 V, 1000 rpm, 4 pole d.c. shunt motor with following design data.
Full load efficiency = 0.83, max flux density in the air gap = 0.9 wb/m², ac = 30,000 A/m, field form factor = 0.7, Max. efficiency occurs at full load, the field current is 2.5% of the rated current, a square pole face is used.

7. With the help of flow chart explain the design of squirrel cage rotor of an induction motor.

OR

With the help of flow chart explain the steps for designing the main dimensions of an alternator.

8. Write short notes (any two).
 - a) Difference between power transformer and distribution transformer.
 - b) Difference between 3 Φ squirrel cage induction motor and slip ring induction motor.
 - c) Algorithms for optimal designs by example.
 - d) Programming techniques (LP & NLP).

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